

C1: Data Quality & Completeness

-  Incomplete data
-  Noisy inputs
-  Reporting bias
-  Missing negatives
-  Inconsistent formats

O1: Standards, Benchmarks & Validation Pipelines

- ✓ Minimal reporting standards
- ✓ Standardized ontologies
- ✓ Benchmark suites
- ✓ Automated curation
- ✓ Anomaly detection

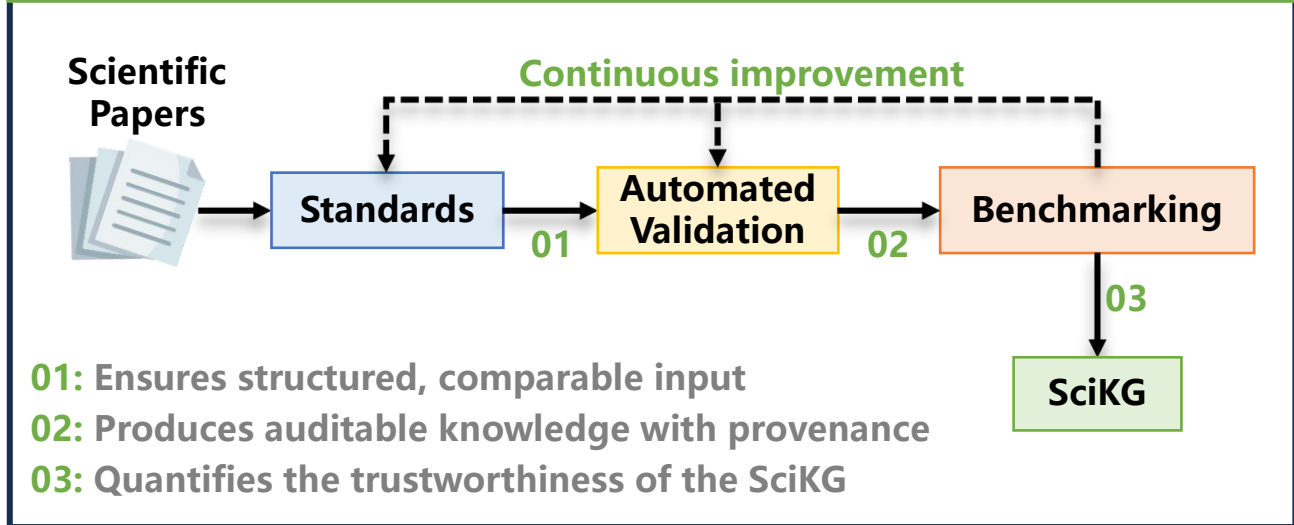
C2: Interoperability & Integration

-  **Domain-specific ontologies**
-  **Proprietary restrictions**
-  **Inconsistent metadata standards**
-  **Siloed development**
-  **Limited cross-disciplinary reasoning**

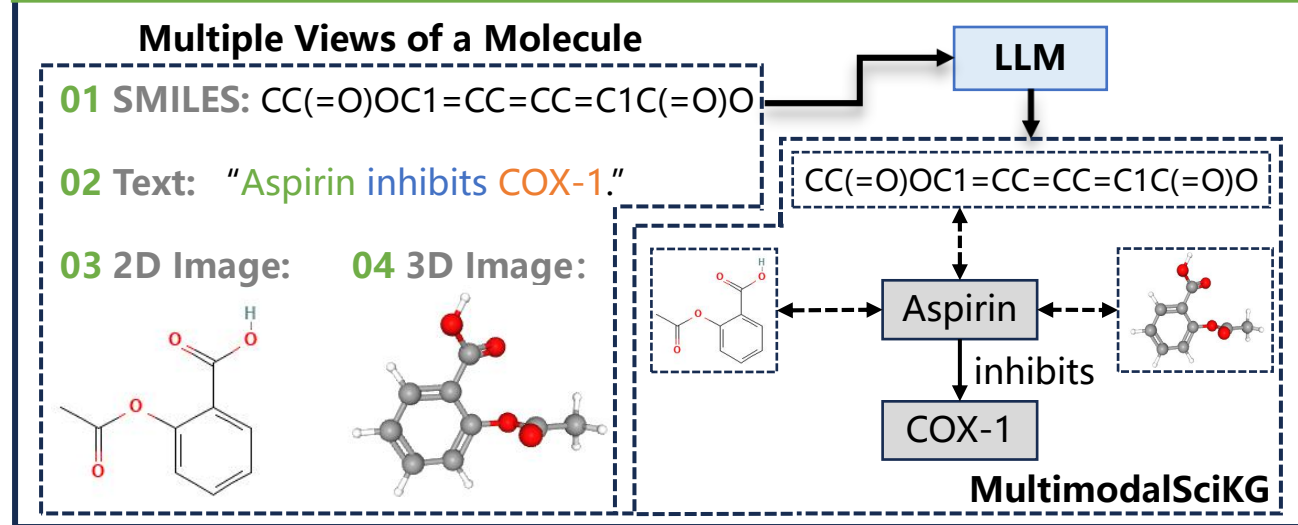
O2: Multimodal Foundation models

- ✓ **Unified ontological frameworks**
- ✓ **Federated knowledge integration**
- ✓ **Cross-modal alignment**
- ✓ **Semantic translation**
- ✓ **Bidirectional knowledge enrichment**

Towards Auditable, Trustworthy SciKGs



Towards Unified, Cross-Modal SciKGs



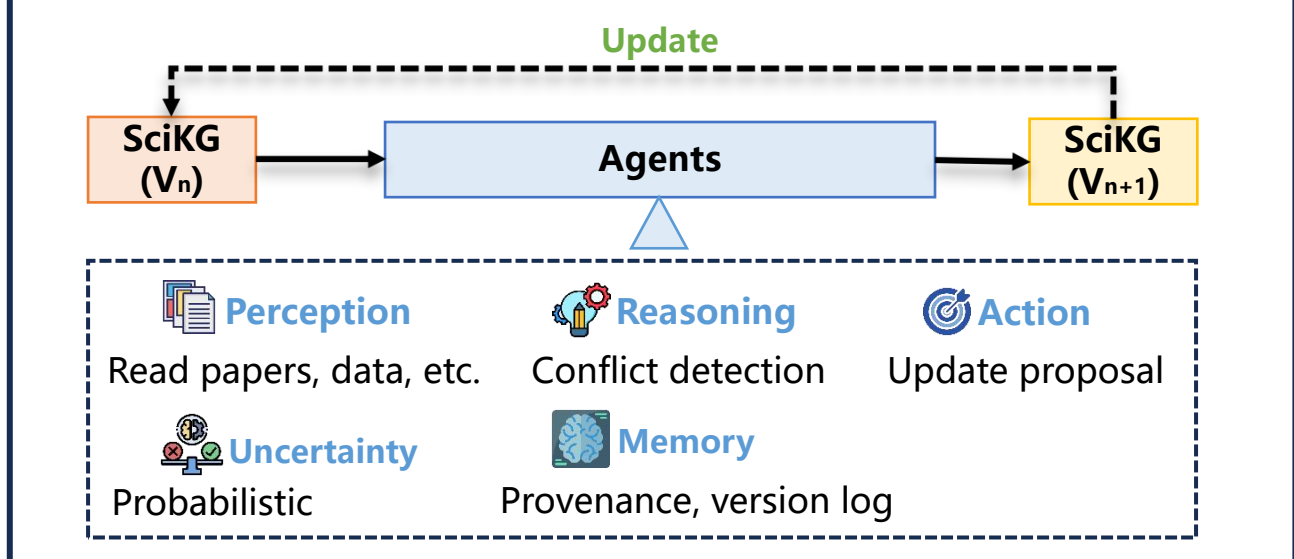
C3: Dynamic & Temporal knowledge

- Static SciKGs
- No version control
- Uncertainty ignored
- Retractions not tracked
- Competing hypotheses

O3: Autonomous Updating via Agents

- ✓ **AI scientific agents**
- ✓ **Continuous monitoring**
- ✓ **Incremental updates**
- ✓ **Provenance tracking**
- ✓ **Confidence scoring**

Towards Dynamic, Autonomous SciKGs



C4: Trustworthy & Explainable Reasoning

- Black-box reasoning
- Bias propagation
- Opaque provenance
- Untraceable conclusions
- Ethical risks

04: Community-Driven Platforms

- ✓ **FAIR principles**
- ✓ **Open governance**
- ✓ **Collaborative curation**
- ✓ **Audit trails**
- ✓ **Explainable modules**

Towards Verifiable, Community-governed SciKGs

